MIL-P-48372 (AR)
27 JULY 1978

MILITARY SPECIFICATION

PROPELLANT, M30A1
FOR USE IN CHARGE, PROOF, 155MM, PXR-6255

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30A1 Propellant for use in one type of propelling charge designated as charge, proof, 155MM, PXR-6255, for 155MM Cannon M185 and M199.

1.2 Classification. The propellant shall by Type 1, cylindrical, multi-perforated grains.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS (see 6.8)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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4.5.6.2.) calculation for average burning rate. The

9289154 - Charge, Proof, 155MM, PXR-6255 for 155MM Cannon M185 and M199

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision)).

AMC PAMPHLET 760-114

Experimental Statistics, Section 5, Tables

(Application for copies should be addressed to the Publications and Reproduction Agency, Letterkenny Army Depot, Chambersburg, Pa. 17201).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M30A1, conforming to the requirements of MIL-STD-652 and Drawing 9289154.

3.2 Ballistic assessment requirements. The propellant, when fired in the M185 or M199 cannon (with 95 percent minimum life remaining tube) and with the projectile (M549 and M483A1) (see 6.9) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.4.1.1. At 70°F (21.1°C) and standard weapon conditions, the propellant shall be capable of assessment to a average pressure of 57,500 pounds per square inch (psi) (396.4 megapascals (MPa)), for the M549 projectile. The test which produces a M549 projectile pressure assessment of 57,500 psi shall produce a corresponding average pressure (59,000 psi informational) when tested with the M483A1 projectile.

3.3 Ballistic uniformity requirements. The propellant, when fired in the M185 or M199 cannon (with 95 percent minimum life remaining tube) at the established charge weight and with the projectile (M549 or M483) (see 6.12) for which



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it was manufactured, shall comply with the following requirements when tested as specified in 4.4.1.2.

3.3.1 Maximum allowable individual pressure. The propellant shall not produce a chamber pressure greater than 59,800 psi (409.5 MPa) for the M549 projectile when fired in a standard weapon.

3.3.2 Data requirements. The contractor shall generate data, in accordance with the requirements of the data item descriptions cited in 4.3.2.1 and 4.3.2.2, documenting specific laboratory and Proving Ground test data.

3.4 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Preproduction Sample Inspection and Quality Conformance inspection shall be conducted on this item.

4.3 Quality conformance inspection

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and testing

4.3.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. The contractor shall submit a Propellant Description Sheet in report form and in computer-compatible format (keypunched cards) in accordance with Data Item Description DI-T-4904, giving the detailed test results (see 6.1.2).



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4.3.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652, except that ten (10) containers shall be sampled for initial production lots and subsequent production lots. The Proving Ground shall submit an acceptance Test Summary in report form and in computer-compatible format (keypunched cards) in accordance with Data Item Description DI-T-4904, giving the detailed test results (see 6.1.2).

4.3.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify DRDAR-QAR-R and DRSAR-QAM, and DRCPM-CAWS-AM.

4.3.2.4 Average pressure (see 3.3.2). If any average pressure obtained at 70°F (21.1°C) exceeds the applicable requirement, apply retest criteria in accordance with 4.4.1.3.

4.3.2.5 Maximum allowable individual pressure (MAIP). If any individual pressure observed during the 70°F (21.1°C) uniformity series, when corrected to standard conditions and recommended charge weight, exceeds the applicable requirement, the lot shall be rejected.

4.4 Test methods and procedures

4.4.1 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedures. The following tolerances shall apply to 70°F (21.1°C) conditioning: 70°F (21.1°C) from 67.5°F (19.7°C) to 72.5°F (22.5°C).

4.4.1.1 Charge establishment. For each of the projectiles, M549 and M483, three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F (21.1°C). From the data obtained, a charge weight shall be calculated which will yield the prescribed pressure (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.4.1.2). The difference between the corrected pressure obtained during the uniformity series and the service pressure shall be compensated for by a final adjustment in charge weight using the slope of the charge weight curve previously established.

4.4.1.2 Uniformity. For the M549 projectile, ten rounds shall be assembled with the established charge weight of propellant and shall be conditioned to 70°F (21.1°C) (as



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indicated by thermocoupled control round). Firing results shall be corrected to standard projectile weight (see 3.2), to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight) and for deviations of the pressure shall be calculated using corrected results.

4.4.1.3 Retest criteria (see Figure 1). If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary (see 6.7).

a. Test lot fails average pressure at 70°F (21.1°C).

b. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.4.1.1.

## 5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A (see 6.11).

## 6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions.

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply.

6.1.2 Contract data requirements. Deliverable data required by this specification is cited in the following paragraphs;

| Paragraph | Data Requirement | Applicable Data Item Description |
|---|---|---|
| 4.3.2.1 | Propellant Description Sheet | DI-T-4904 |
| 4.3.2.1 | Keypunched Cards | DI-T-4904 |
| 4.3.2.2 | Acceptance Test Summary | DI-T-4904 |
| 4.3.2.2 | Keypunched Cards | DI-T-4904 |

One copy of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to:



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Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR
DRDAR-LCU-EP, DRCPM-CAWS-AM
Dover, New Jersey 07801

One copy of each set of keypunched cards shall be forwarded, ATTN: DRDAR-QAR-R. (Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Surveillance sample. Samples shall be forwarded to US Army R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.4 Test rounds. Test rounds shall be assembled in accordance with drawing 9289154.

6.5 Proving ground test summary (see Table I).

6.6 Retest. All original and retest data shall be submitted to US Army R&D Command, ATTN: DRDAR-QAR-R, & DRCPM-CAWS-AM Dover, New Jersey 07801.

6.7 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the congizance of ARRADCOM.

6.9 Projectile weight. The average projectile weight for the propelling charge in which this propellant will be loaded will be determined by the loading plant and provided for information by DRSAR-QAM to the Proving Ground. The actual average projectile weight will be used for charge weight



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assessment of propellant. All projectiles intended for use in test rounds shall be inert-loaded to a tolerance of ± .01 lb. (4.5 gm). The Loading Authorization for the propellant lot should be annotated to indicate that the service charge weight listed is to be used only when loading with projectiles having the listed weight. If the projectile weight is different from that listed, the lot should be reassessed to the new projectile weight or the charge weight adjusted using a known projectile weight/charge weight curve. Calibration projectile weights shall be in accordance with Dwg. 9289154 except that a tolerance of ± .01 lb. (4.5 gm) shall apply.

6.10 Propellant Sample. Whenever a propellant lot is selected for use, a 5-lb sample shall be forwarded to US Army Armament R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-QAR-R for use in evaluating the propellant stockpile.

6.11 Intended Use. The proof charge covered by this specification is intended for use with 155MM projectile development and acceptance tests and for M185 and M199 Cannon Proof Tests. This charge is intended for Proving Ground use only to be fired in a M199 or M185 Cannon with all personnel behind a protective barrier. This charge is not intended for firing when the charge temperature is less than 35 degrees Fahrenheit or greater than 75 degrees Fahrenheit.

6.12 Test Matrix. (See Figure 1).

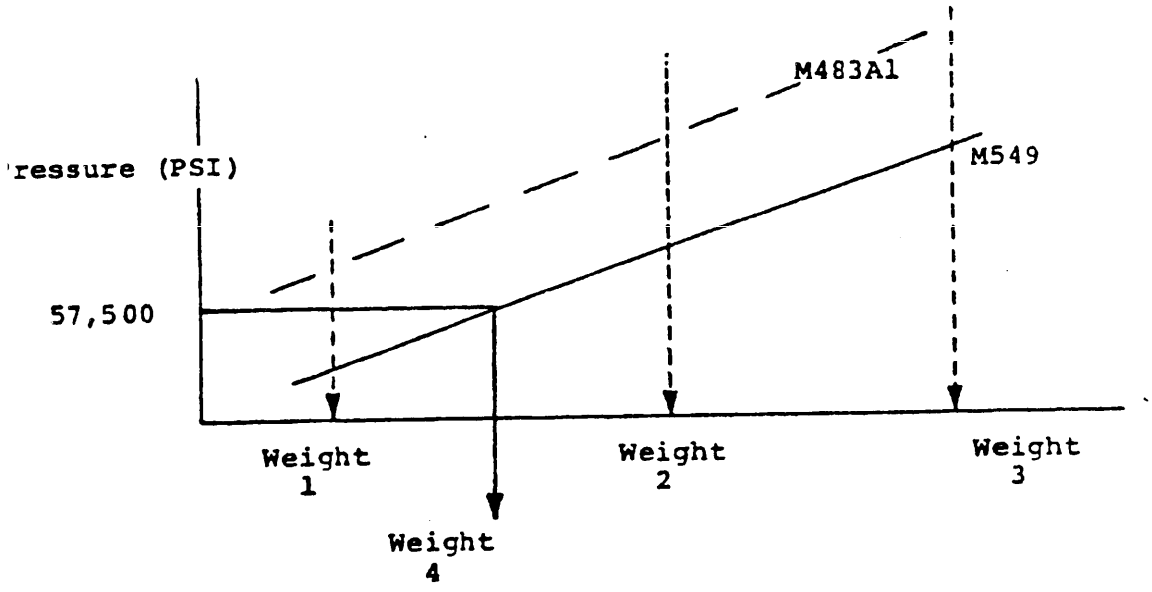



Figure 1
TEST MATRIX



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During charge establishment 6 charges at each weight 1, 2, and 3 shall be loaded. Three charges at each weight shall be fired with the M549, and the three with the M483A1 to determine the charge slopes. The slope of the M549 pressure weight curve shall be used to determine the propellant weight (Weight 4) at 57,500 psi pressure. The corresponding pressure on the M483A1 slope shall be determined after the M549 ten round uniformity test.

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TABLE I

PROVING GROUND TEST SUMMARY

| Temperature (°F) | Temperature (°C) | Zone | Test | SAMPLE SIZE | TEST RESULTS | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Average Vel fps mps | Std Dev of Press psi MPa | Max Allowable Individual Pressure psi MPa | | Maximum Average Pressure psi MPa | | P/T Trace | Differential Pressure |
| CHARGE ESTABLISHMENT | | | | | | | | | | | |
| 70 | 21.1 | N/A | 9 (M549) | Info. | N/A | N/A | | 57,500 | 396.4 | Info. | Info. |
| | | | 9 (M483A) | Info. | N/A | N/A | | 59,000 | 406.8 | Info. | Info. |
| UNIFORMITY (INITIAL PRODUCTION) | | | | | | | | | | | |
| 70 | 21.1 | N/A | 10 (M549) | Info. | Info. | 59,000 | 406.8 | 57,500 | 396.4 | Info. | Info. |



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MIL-P-48372 Propellant M30A1 For Use In Charge Proof 155MM, PXR-6255

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